Atmospheric Research 250 (2021) 105365

Contents lists available at ScienceDirect

Atmospheric Research

journal homepage: www.elsevier.com/locate/atmosres

# Spatial performance of multiple reanalysis precipitation datasets on the southern slope of central Himalaya

Yingying Chen [a,b,*], Shankar Sharma [a,c], Xu Zhou [a], Kun Yang [a,d], Xin Li [a,b], Xiaolei Niu [a], Xin Hu [a,c], Nitesh Khadka [c,e]

[a] National Tibetan Plateau Data Center, Key Laboratory of Tibetan Environmental Changes and Land Surface Processes, Institute of Tibetan Plateau Research, Chinese Academy of Sciences, Beijing 100101, China
[b] CAS Center for Excellence in Tibetan Plateau Earth Sciences, Beijing 100101, China
[c] University of Chinese Academy of Sciences, Beijing 100049, China
[d] Ministry of Education Key Laboratory for Earth System Modeling and Center for Earth System Science, Tsinghua University, Beijing 100084, China
[e] Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, Chengdu 610041, China

ARTICLE INFO

*Keywords:*
Precipitation
Spatial pattern
Reanalysis datasets
ERA
HAR
Nepal

ABSTRACT

The performance of gridded precipitation products is especially important over mountainous regions because of the lack of observations. This study provides a detailed evaluation of the spatial precipitation patterns presented by the ERA-Interim, ERA5, ERA5-Land, and refined HAR datasets with spatial resolutions from coarse to fine (0.7°, 0.25°, 0.1°, and 0.1°, respectively) over the southern slope of central Himalaya, Nepal. The major findings are as follows: (1) The high-resolution ERA5-Land and ERA5 datasets well present the observed spatial pattern of precipitation but with a generally overestimated amount; the refined HAR dataset also presents many fine-scale details, while the coarse resolution ERA-Interim dataset tends to exhibit a too gentle spatial characteristic in mountainous Nepal. (2) The four datasets well reproduce the seasonal precipitation variation ($R > 0.90$) over the country; however, their performance varies spatially across the western, central, and eastern regions. The spatial resolution is also important in representing the terrain blocking of water vapor from the Bay of Bengal, which is transported from east to west in Nepal. (3) The high-resolution datasets can reproduce the elevation dependency as revealed by the observed dataset, while the coarse-resolution cannot capture the elevation dependency in the high-elevation regions. Moreover, all four datasets show very similar performance in low–elevation plane areas, however, higher resolution simulations (ERA5, ERA5–Land, and HAR) better reflect blocking effect of orographic precipitation and outperform ERA-Interim in high–elevation areas where the terrain is very complex and the slope is very steep.

## 1. Introduction

Precipitation is a controlling factor in the water cycle and the most active element associated with atmospheric circulation in weather and climate (Trenberth et al., 2003; Kidd and Huffman, 2011). In addition, precipitation is also one of the most challenging variables to predict or estimate (Tapiador et al., 2012). Reliable and accurate precipitation data are very important for weather and climate studies, the management of water resources and hydrological forecasting, which require proactive measures to investigate natural hazards (Jiang et al., 2012; Sarojini et al., 2016; Liu et al., 2017; Faiz et al., 2018).

Nepal is located between the northern part of the Indo-Gangetic plains and the southern part of the main Himalayan mountain ranges. The abrupt changes in topography with different climatic zones and weather patterns within a short latitudinal range in Nepal challenge the precise observation of meteorological parameters (temperature, precipitation, and humidity) throughout the country. Surface gauge-based observations (hereafter referred to as "OBS") are the only direct measurements that provide highly precise measurement of rainfall. However, the sparse distribution of gauges across some regions with highly varying topography remains a major issue in the representation of accurate precipitation (Bhardwaj et al., 2017; Sun et al., 2018). For

* Corresponding author at: National Tibetan Plateau Data Center, Key Laboratory of Tibetan Environmental Changes and Land Surface Processes, Institute of Tibetan Plateau Research, Chinese Academy of Sciences, Beijing 100101, China.
*E-mail address:* chenyy@itpcas.ac.cn (Y. Chen).

https://doi.org/10.1016/j.atmosres.2020.105365
Received 9 September 2020; Received in revised form 25 October 2020; Accepted 11 November 2020
Available online 12 November 2020
0169-8095/© 2020 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

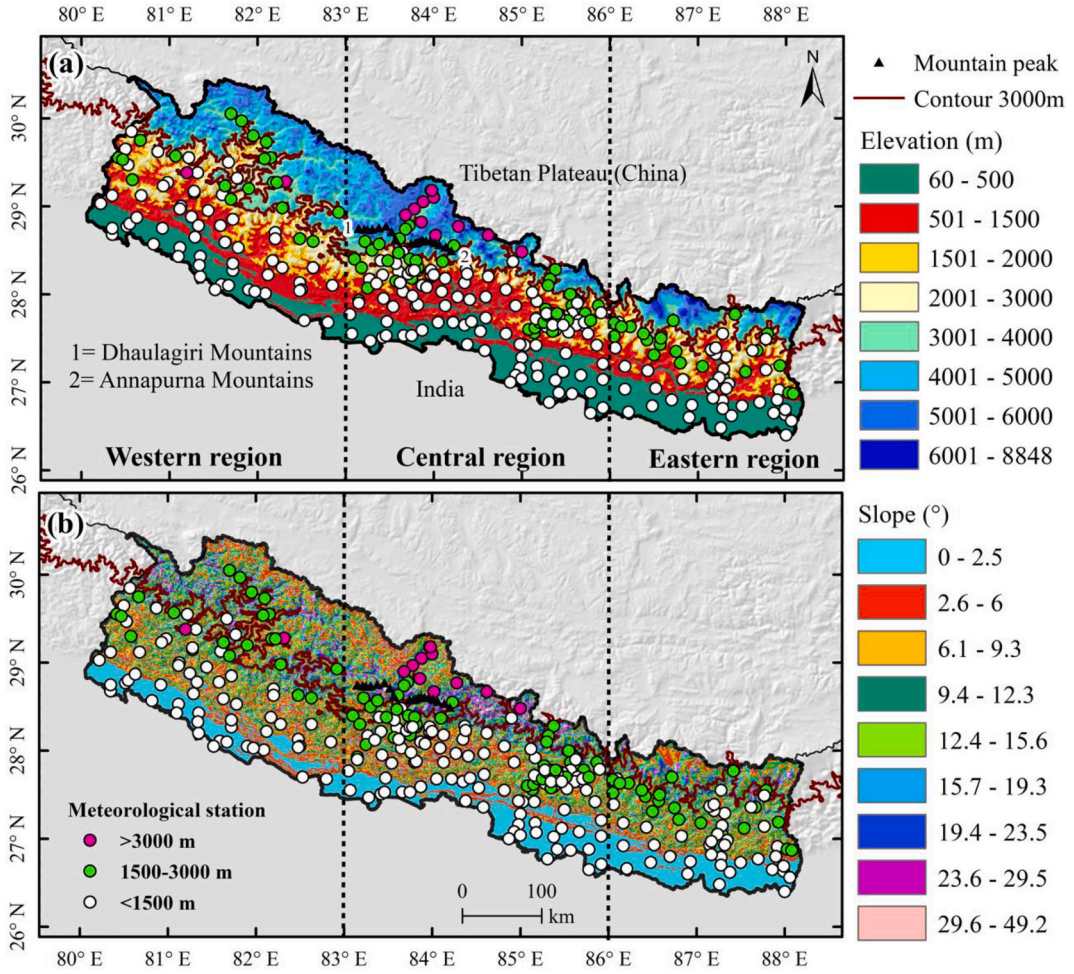



**Fig. 1.** Location of selected meteorological stations across Nepal with (a) the digital elevation model and (b) slope (°) of the topography as the background. White, green, and cyan dots denote the stations located below 1500 m, between 1500 and 3000 m and above 3000 m, respectively. The black and black dotted lines are the national border and the subregion boundaries, respectively.

example, in Nepal, fewer stations are found in high-elevation regions than in low-elevation regions (Lang and Barros, 2002; Barros and Lang, 2003; Barros et al., 2006). The country experiences several extreme precipitation events annually (Talchabhadel et al., 2018), which often lead to natural disasters, such as landslides (van der Geest, 2018) and floods (Delalay et al., 2018); reliable long-term precipitation datasets are therefore needed to investigate such processes.

Reanalysis and refined datasets of high spatial and temporal resolution are potential alternatives to rainfall measurements in station-sparse regions (Essou et al., 2017; Sahlu et al., 2017) for regional meteorological and climatic studies. These datasets produce a coherent set of surface and upper air circulation data (Kalnay et al., 1996; Bosilovich et al., 2013; Hu et al., 2016). The ERA-Interim (hereafter referred to as "ERA-Int") (Dee et al., 2011) and its updated versions, i.e., the ERA5 (Hersbach and Dee, 2016) and ERA5-Land (hereafter referred to as "ERA5-L") (C3S, 2017) datasets, are global reanalysis datasets produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). In addition to the global reanalysis datasets, the regional refined dataset, e.g., the High Asia Refined analysis (hereafter referred to as "HAR") (Maussion et al., 2014), is produced through the dynamical downscaling of global reanalysis data by regional climate models for the Tibetan Plateau and the surrounding region. The reanalysis and regional refined datasets include many climatic state variables, making them ideal for investigating climate variability and for improving water resource management. However, the spatial resolution, local climate and topography can influence the performance of these reanalysis and refined datasets. Thus, independent evaluations of these datasets are required to determine their tendencies and discrepancies in representing precipitation patterns.

Gridded precipitation products have been evaluated and used for many different applications worldwide (Lindsay et al., 2014; Li et al., 2017; Yang et al., 2017; Lu and Yong, 2018; Kolluru et al., 2020; Sharma, 2020; Wang et al., 2020a; Yao et al., 2020). The precipitation pattern in a mountainous region is inherently complex because of interactions between the terrain and atmosphere. Previous studies have investigated the performance of reanalysis datasets in various aspects, globally. For example, Ceglar et al. (2017) evaluated four reanalysis datasets (ERA-Int, ERA-Int/Land, AgMERRA, and JRA-55) with two gridded observational datasets (APHRODITE and Climate Hazards Group Infrared Precipitation with Stations (CHIRPS)) over monsoon Asia (0°–40°N, 65°–125°E). They found that AgMERRA achieves the best overall performance among the datasets. Rana et al. (2015) analyzed the performance of satellite and reanalysis (ERA-Int and CFSR) based precipitation datasets against rain gauge observation over the Indian subcontinent and found that these reanalysis datasets can reproduce seasonality in low-elevation regions, but uncertainties remained in the mountainous region. Recently, Tarek et al. (2019) and Xu et al. (2019) reported that the uncertainties in ERA5 precipitation were consistently smaller than those of ERA-Int for evaluation over North America and the Assiniboine River Basin (Canada-United States border), respectively. An evaluation of the gridded reanalysis precipitation datasets in the Pamir mountains of central Asia shows that higher

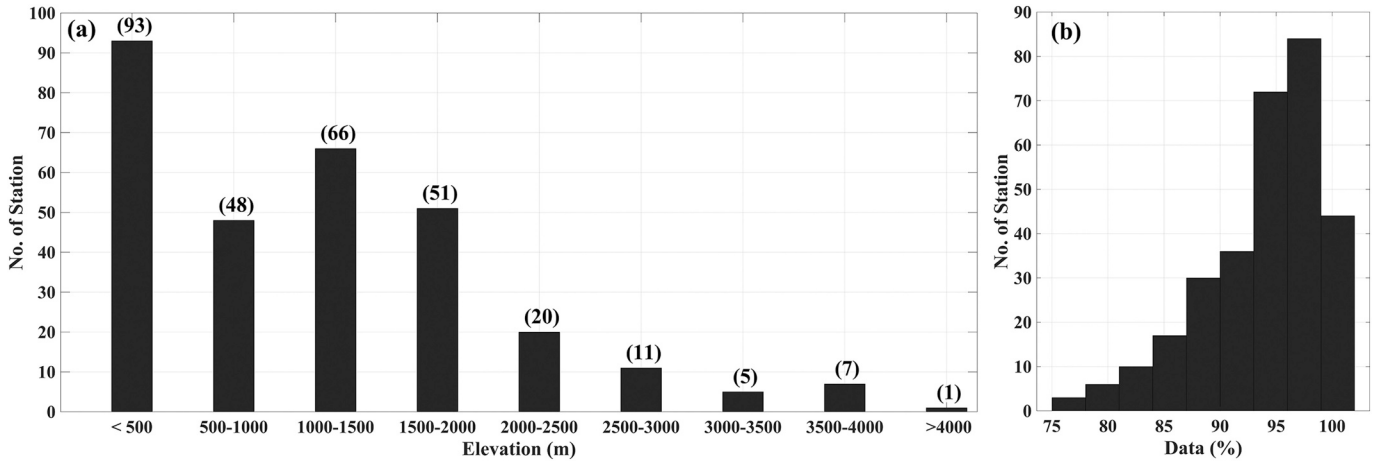



**Fig. 2.** The number of stations at different elevation ranges (a), and histogram for number of stations with data availability (%) during the study period of 2001–2013 (b).

resolution reanalysis datasets substantially improved the performance over those of coarse resolution datasets (Zandler et al., 2019). Meanwhile, in the central and eastern Himalayas of Nepal, Menegoz et al. (2013) analyzed the precipitation and snow cover using the MAR (Modèle Atmosphérique Régional) model. The authors only compared the atmospheric circulation described in the reanalysis with the model. They found that a regional climate model with a high spatial resolution is very useful for depicting the terrain more realistically and describes the effect of orography more accurately than coarse resolution. Li et al. (2020) also found that high-resolution datasets (HAR10 and HAR30) can reasonably represent the amount of precipitation in the high mountain regions of the western Tibetan Plateau. In short, most global studies, as well as regional studies, concluded that further investigation is still needed to analyze the effect of the orographic effect in precipitation simulation, especially in complex terrains.

Comprehensive evaluations of precipitation data from reanalysis datasets are essential, especially in regions where the terrain is very rugged (e.g., Nepal). Recently, Hamal et al., 2020 evaluated the MERRA-2 reanalysis precipitation product and found that the gauge calibrated MERRA-2 product shows a more consistent spatial performance compared to that of model-only data in Nepal. However, the author did not discuss the model resolution and its effect on the precipitation simulation in the mountainous region. Further, spatial resolution plays an important role in representing an orographic gradient and moisture transfer especially in mountainous regions of Nepal. Moreover, the performance of the newly developed high-resolution reanalysis and refined datasets in representing the spatial pattern of precipitation in the country has not been reported yet. The complex topographic nature of Nepal, i.e. elevation rises from 60 m in the south to 8848 m at the summit of Everest within the 200 km width of the country, creates the heterogeneous precipitation distribution. Thus, it is very important to evaluate the role of spatial resolution in representing the spatial pattern of precipitation in this region, as these factors affect the accuracy of the reanalysis and refined products.

This study aims to fill this research gap by evaluating the precision of precipitation data from the widely used ERA-Int, the recently developed ERA5 and ERA5-L, and the high-resolution refined HAR, focusing on the role of spatial resolution in representing the effects of an orographic gradient on a precipitation simulation in a mountainous region using 302 gauges observation of Nepal. The main objectives of the study are to deliver a detailed quantification of the error characteristics and spatial accuracy of the above four datasets in Nepal. Furthermore, this study will be useful for providing guidelines and important references for choosing appropriate datasets for future scientific research, which is very important in complex terrain regions and station sparse regions.

The remainder of this paper has the following structure: Section 2 describes the data and methodology, Sections 3 and 4 present the results and discussion, respectively, and in Section 5, we summarize the study.

## 2. Data and methodology

### 2.1. Study area

Nepal is located within the geographic region of 26.36°–30.45°N and 80.06°–88.2°E. The country covers 147,516 km$^2$, with abrupt changes in altitude ranging from 60 m to 8844 m within a 150–200 km distance (Fig. 1). With these unique physiographical and topographical distributions, the country features a variety of climates that range from the tropical savannah over the southern plain to the polar frost in the northern mountains (Karki et al., 2016). Nepal has four climatological seasons: the pre-monsoon season (March-May), summer monsoon season (June-September), post-monsoon season (October-November) and winter season (December-February) (Nayava, 1980). The wet air masses are transported from the Bay of Bengal and occasionally from the Arabian Sea, causing widespread precipitation. The summer monsoon reaches eastern Nepal in mid-June and advances westward, covering the entire country within a week (Nayava, 1980; Shrestha, 2000). The South Asian monsoon contributes 97% of the annual precipitation during the monsoon seasons, while the remaining 3% of the precipitation occurs in the winter because of the influence of the westerlies system (Hamal, 2020). The average annual precipitation in Nepal is ~1500 mm, and it varies among locations according to the geographical features and is fundamentally controlled by the monsoon system (Sharma et al., 2020). As monsoon starts from eastern side of the country, center part receives highest amount of monsoonal precipitation, while lowest in western part. In contrast, winter precipitation is more pronounced in western hills than central and eastern part due to the influence of western disturbances. Those differences in precipitation seasonality and pattern within three major river basins suggest that the country can be divided into three distinct regions, i.e., western (western boundary to 83°E), central (from 83° to 86°E), and eastern region (from 86°E to eastern boundary) (Black dotted line in Fig. 1). Further, these three different regions feature different monsoonal and winter climatology, and it is a common practice to study the precipitation climatology in the country (Pokharel et al., 2019; Sharma, 2020), which is also suggested by Kansakar et al. (2004).

### 2.2. Data sources

#### 2.2.1. Observed data

There are currently ~549 precipitation network stations established in Nepal, but not all these stations make regular observations. This study selected daily precipitation data from a total of 302 gauge stations from January 2001 to December 2013. Most of these observations use the same type of US standard 8-in. diameter manual precipitation gauges (Goodison et al., 1998), which are very consistent in terms of measurement method (Talchabhadel et al., 2017). The spatial distribution of the stations varies considerably throughout the county, with an approximate average density of two stations per 1000 $km^2$, and ~ 90% of the stations are located below 3000 m in elevation (Fig. 1a). Because of the remote locations and difficulty of routine maintenance, only a few stations cover the high-elevation, mountainous regions of the county. The number of stations at different elevation intervals are presented in Fig. 2a. Because of the lack of automatic gauge stations in high-elevation regions, the Department of Hydrology and Meteorology (DHM) station only measures the total precipitation (rainfall and snowfall) amount. All these observed data were collected from the DHM (www.dhm.gov.np), Government of Nepal. All these stations do not feature regular observations especially in high-elevation areas due to harsh environment (during winter) and remote location to routine maintenance. However, we have selected the maximum number of stations featuring more than 75% daily data with optimal spatial coverage. The data availability (in %) in selected 302 stations during the study period in Fig. 2b.

#### 2.2.2. Reanalysis and refined datasets

ERA-Int is a global atmospheric reanalysis project developed by the ECMWF that provides an improved atmospheric model and assimilation system that replace those used in ERA-40 (Dee et al., 2011). The dataset dates back to 1979 and is updated with an approximately one-month delay from real-time. The ERA-Int 6-hourly total precipitation data with a horizontal resolution of 0.70° × 0.71° were downloaded from (https://apps.ecmwf.int/datasets/data/interim-full-mnth/levtype=sfc/, accessed on December 2019) for the study period. The ERA-Int data are produced using the Four-Dimensional Variational (4D-Var) assimilation in CY31R2 (model–cycle) of the Integrated Forecast System (IFS). 4D-Var assimilation technique is a statistical interpolation in space and time between a distribution of meteorological observations and an a prior estimate of the model state (the background). More information on these data is available at the ECMWF reanalysis site (https://www.ecmwf.int/en/forecasts/documentation-and-support).

ERA5 was recently developed as the fifth generation of the European reanalysis products generated using the 4D-Var data assimilation in CY41R2 of the ECMWF IFS (Hersbach and Dee, 2016). ERA5 has 137 hybrid sigma/pressure (model) levels in the vertical and a top-level at 0.01 hectopascal. ERA5 uses one of the latest versions of the ECMWF Earth system model data assimilation methods, which uses an updated parameterization of Earth processes compared to the older model in ERA-Int and is therefore expected to replace ERA-Int. Compared with the earlier ERA-Int, two major changes were implemented in ERA5: the temporal resolution was increased from 3-hourly to hourly and the spatial resolution was increased from 79 km to 25 km (Albergel et al., 2018). The ERA5 and ERA-Int datasets used in this study are forecast variables that did not assimilate the observed precipitation datasets. Hourly ERA5 forecast precipitation data (total precipitation) for the study period with a spatial resolution of approximately $0.25° \times 0.25^0$ were downloaded. More information is available at the ECMWF reanalysis site (https://confluence.ecmwf.int//display/CKB/ERA5±data± documentation, accessed on December 2019).

ERA5-L has been produced by replaying the land component of the ERA5 climate reanalysis and covers the same period as ERA5, January 1950 to near real-time (NRT). ERA5-Land is produced under a single simulation, without coupling to the atmospheric module of the ECMWF's IFS. Observations indirectly influence the simulation through the atmospheric forcing of ERA5. This forcing drives the ERA5-Land single simulation and it has been obtained by assimilating observations through a 4D-Var data assimilation system and a Simplified Extended Kalman Filter (C3S, 2019). The core of ERA5-L is the Tiled ECMWF Scheme for Surface Exchanges over Land incorporating land surface hydrology (H-TESSEL). In particular, ERA5-L runs at an enhanced resolution of ~10 km as compared to ~25 km in ERA5. The temporal frequency of the output is hourly, and the fields are masked for all oceans. The hourly total precipitation data for the study period with a spatial resolution of approximately $0.10° \times 0.10^0$ were downloaded. More information is available at the CDS Climate Copernicus website (https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-land?tab=overview, accessed on February 2020).

The HAR dataset is generated by the dynamical downscaling of global analysis data (Final Analysis data from the Global Forecasting System; dataset ds083.2, a product of the Global Data Assimilation System (GDAS)) using the Weather Research and Forecasting (WRF-ARW) model (Maussion et al., 2014). The HAR dataset did not assimilate observations during the simulation time. The quality of the dataset is strongly dependent on the quality of the global input data and the capacity of the WRF model to simulate atmospheric processes. The HAR dataset provides gridded fields, such as those of the temperature, precipitation, and wind, at a 30 km resolution for central Asia and 10 km resolution for the Tibetan Plateau and surrounding regions for October 2000 to October 2014. For this study, HAR daily total precipitation datasets from January 2001 to December 2013 with a horizontal resolution of 0.10° × 0.10° were used. More information is available at (https://www.klima.tu-berlin.de/index.php?show=forschung_asien_tibet_har&lan=en, accessed on December 2019). Table 1 provides a summary of the above datasets used in this study.

**Table 1**
Overview of the datasets included in this study with the corresponding spatial resolution, temporal coverage, evaluation time scale, and references.

| Datasets | Spatial resolution | Temporal coverage | Evaluation time scale | Reference |
|---|---|---|---|---|
| DHM | 302 Stations | 2001–2013, Daily | Monthly, seasonal | (DHM, 2017) |
| ERA-Int | 0.70° × 0.71° | 2001–2013, 6-hourly | Monthly, seasonal | (Dee et al., 2011) |
| ERA5 | 0.25° × 0.25° | 2001–2013, 1-hourly | Monthly, seasonal | (Hersbach and Dee, 2016) |
| ERA5-L | 0.1° × 0.1° | 2001–2013, 1-hourly | Monthly, seasonal | (C3S, 2019) |
| HAR | 0.1° × 0.1° | 2001–2013, daily | Monthly, seasonal | (Maussion et al., 2014) |

### 2.3. Methodology

#### 2.3.1. Quality control

The precipitation data from reanalysis and HAR are gridded data, and the OBS data are in the station scale. To compare the station-based OBS data against the gridded precipitation datasets, a station-to-grid comparison was performed in this study to avoid the additional error from gridding the OBS data. The precipitation values from each rain gauge and the grid where the same gauge is located were extracted in pairs for evaluation. This method has been adopted to evaluate gridded datasets with gauge observation by several studies (Feidas, 2010; Li et al., 2013; Nkiaka et al., 2017). Section 4 For each station, daily datasets are computed from the sub-daily ERA-Int, ERA5, and ERA5-L datasets to match the same time scale (daily) as that of the OBS datasets. Further, these reanalysis and refined precipitation rates were first aggregated to the same period to match the measurement time windows (03UTC) as DHM gauge observation (Talchabhadel et al., 2017). All station data do not feature regular data; thus, quality control was conducted for data consistency; if the daily observed data contained a

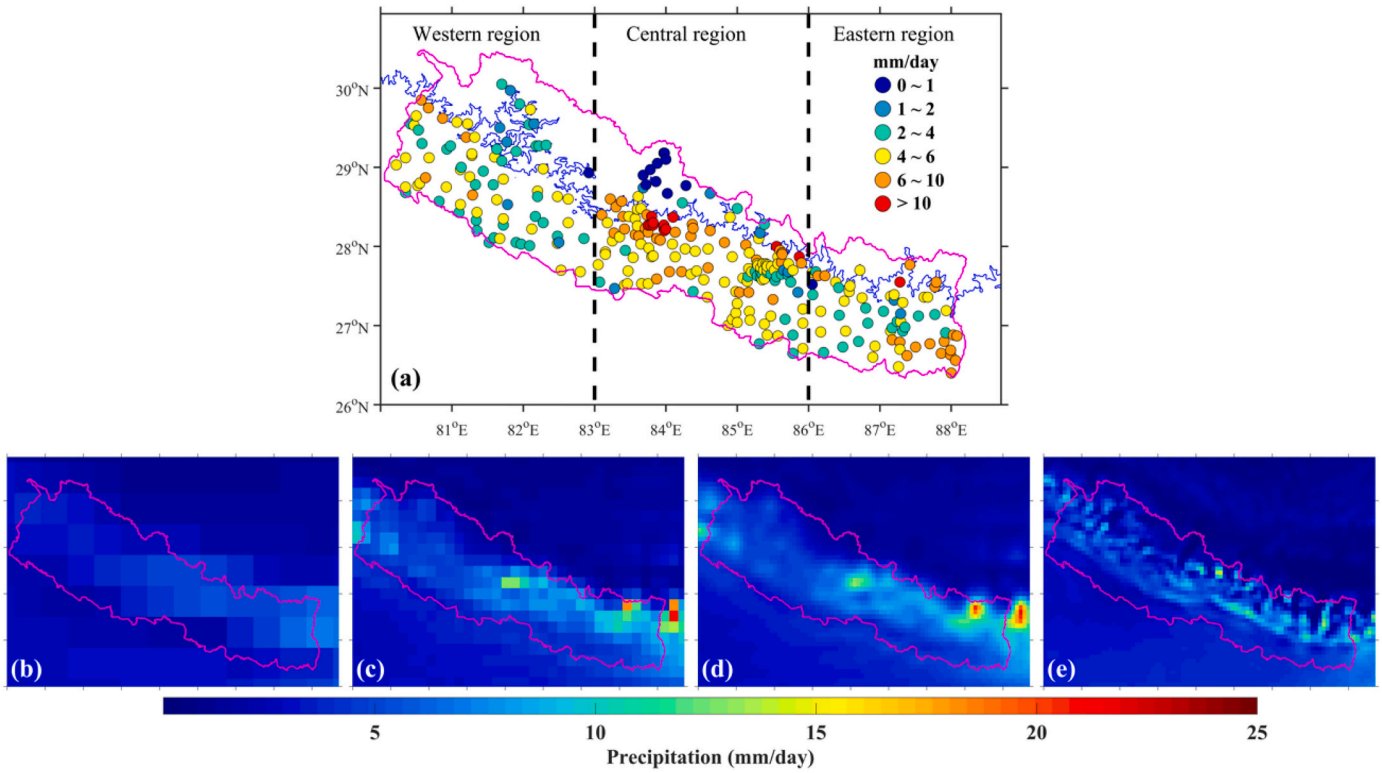



**Fig. 3.** Spatial distribution of the mean precipitation (mm/day) in (a) OBS, (b) ERA-Int, (c) ERA5, (d) ERA5-L, and (e) HAR, averaged over 2001 to 2013. The magenta and black dotted lines represent the national borders and the three subregions, respectively.

missing value, then the corresponding daily data point from the reanalysis and refined datasets was also considered a missing value. Meanwhile, monthly data were computed when a station had more than 25 days of precipitation data available in one month; the precipitation values were converted into a monthly mean; otherwise, the precipitation in that month was considered a missing value. The monthly precipitation data were also considered a missing value from the reanalysis and refined datasets for consistency if the corresponding monthly data were missing from the OBS datasets.

### 2.3.2. Methods

To analyze the spatial distribution of precipitation, the mean precipitation at each station and the mean precipitation from the reanalysis and refined datasets at each grid were calculated for the study period. The mean annual and mean monsoonal (JJAS) precipitation values of the reanalysis, refined, and OBS datasets were calculated for each station. Furthermore, the average annual precipitation variation in these datasets was compared for three regions (i.e. western, central, and eastern regions) (Fig. 1a). To quantify the performance of datasets at different elevations, all datasets (OBS, reanalysis, refined) were divided into six elevation categories every 500 m (i.e., from 60 m to >5000 m). The average slope was also calculated at each of these elevation categories, and a graphical plot of the elevation dependency of the precipitation along with the slope was made to show the variation in precipitation with average slope. The elevation and slope value were computed using 30 m STRMDEM in ArcGIS 10.5. Approximately 31% of the 302 stations were located in the regions below 500 m, and the number of stations decreased with increasing elevation.

Following the previous studies (e.g. Bhardwaj et al., 2017; Fallah et al., 2019; Sharma, 2020), four pairwise statistical metrics were used to quantify the similarities or discrepancies between the OBS and the estimated precipitation data from the reanalysis and the refined datasets (Eqs. (1)–(4)): the correlation coefficient (R, Eq. (1)), which measures the strength and direction of the linear association between datasets; the root mean square error (RMSE, Eq. (2)), which measures the averaged magnitude of the deviation that a dataset has from the OBS; and the mean bias (MB, Eq. (3)) and relative bias (RB, Eq. (4)) (RB is the absolute ratio of the mean bias and observed mean precipitation), which measure any persistent overestimation or underestimation bias of a dataset and the difference between the estimated precipitation and the OBS precipitation. For the spatial performance, the mean monsoonal precipitations at each station were compared in three elevation intervals, while the regional average (western, central, and eastern) of the monthly mean precipitation were compared during the study period. Graphical plots and statistical metrics were used to quantify the performances of the reanalysis and refined datasets. In addition, a graphical plot of the summer monsoon season was made to show the variation in precipitation with latitude. The formulas of the statistical metrics are as follows:

$$R = \frac{\sum_{i=1}^{n}\left(O_i - O^-\right)\left(E_i - E^-\right)}{\sqrt{\sum_{i=1}^{n}\left(O_i - O^-\right)^2}\sqrt{\sum_{i=1}^{n}\left(E_i - E^-\right)^2}} \quad (1)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(E_i - O_i)^2}{n}} \quad (2)$$

$$MB = \frac{\left(E^- - O^-\right)}{n} \quad (3)$$

$$RB = \left|\frac{MB}{O^-}\right| \quad (4)$$

where $O$ is the OBS dataset, $E$ is the estimated precipitation of the reanalysis and refined datasets, $\overline{O}$ and $\overline{E}$ denotes the average value of their respective datasets, and $n$ is the sample size.

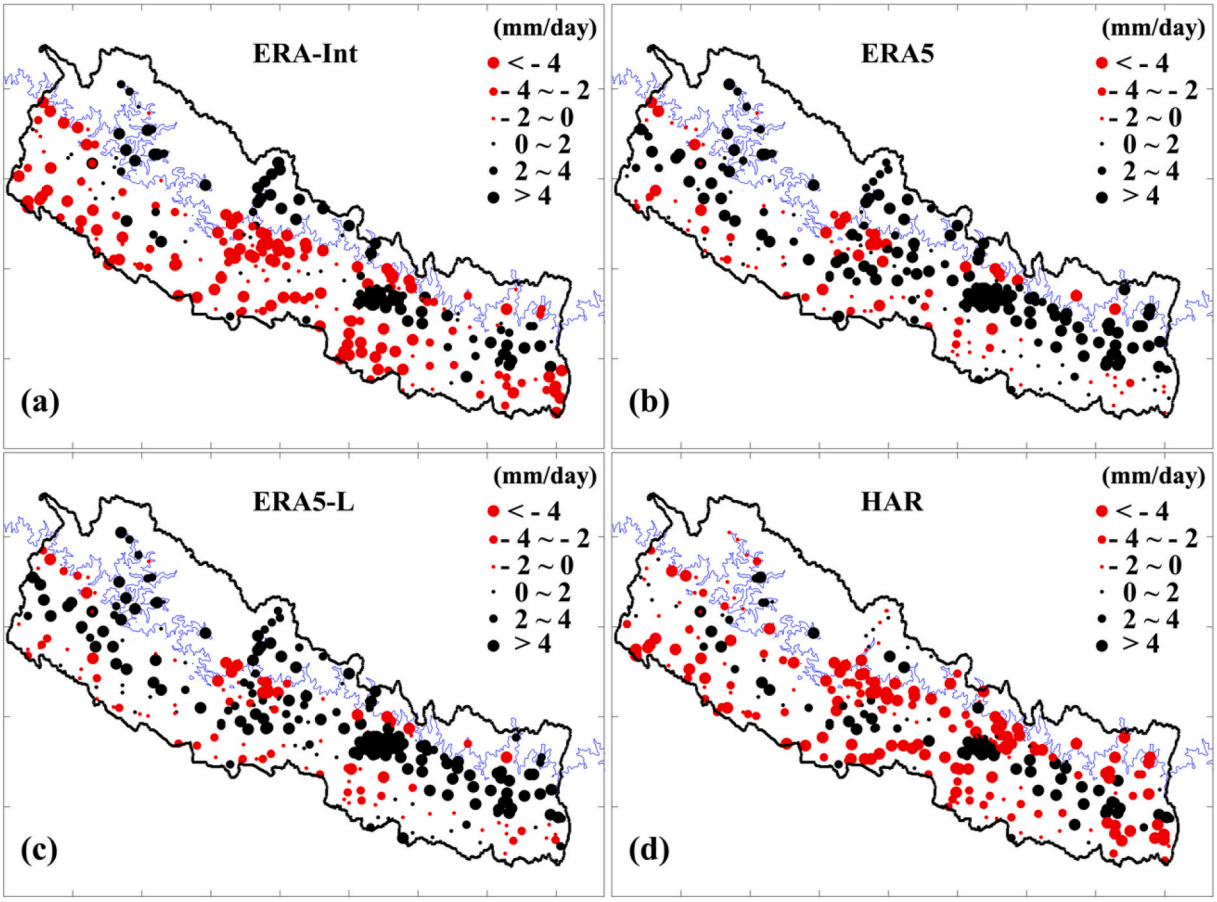



**Fig. 4.** Mean precipitation biases (mm/day) (MB) during the summer monsoon season (JJAS) in Nepal for (a) ERA-Int, (b) ERA5, (c) ERA5-L, and (d) HAR during the study period. Black and blue lines denote the national border and the 3000 m elevation contour line, respectively.

## 3. Results

### 3.1. Spatial distribution of precipitation over Nepal

The spatial distributions of the mean precipitation (mm/day) derived from OBS, reanalysis, and the refined datasets during the study period are presented in Fig. 3. The distribution of the precipitation varies over the country for ERA-Int, ERA5, ERA5-L, and HAR, with the spatial resolution from coarse to fine (0.7°, 0.25°, 0.1°, and 0.1°, respectively). Fig. 3a indicates that the region with the highest observed precipitation amount (above 10 mm/day) is located near 28.3° N, 84° E (with an elevation between 1500 and 2500 m) in central Nepal, i.e., the windward sides of the Annapurna Mountains; while the region with the lowest precipitation amount (~1 mm/day) is located to the north of central Nepal (with an elevation above 2500 m), i.e., the leeward slope of the Dhaulagiri Mountains. This observed precipitation pattern was well captured by the high-resolution ERA5 and ERA5-L reanalysis datasets but with a higher precipitation amount. Notably, the ERA5 and ERA5-L datasets showed another precipitation center on the windward slope of the central Himalayan Mountains in the eastern region of Nepal (Fig. 3c and d), near $27.8^0$N, 87.2°E, with the highest precipitation approximately 18 mm/day and approximately 22 mm/day, respectively. In that region, only two DHM stations were deployed, which is too sparse to represent the regional precipitation center, although they have large precipitation amounts (>6 mm/day). In comparison with the ERA5 and ERA5-L datasets, the coarse resolution ERA-Int dataset failed to present the observed precipitation pattern in Nepal and exhibited a too gentle spatial characteristic. As for another high-resolution dataset, HAR appeared to present many fewer fine-scale details than ERA5 and ERA5-L, which hid the observed spatial pattern of precipitation in Nepal. These results confirm that in the mountainous regions with a complex topography, the spatial resolution plays an important role in representing the effects of an orographic gradient on a precipitation simulation. The importance of the spatial resolution in high-elevation regions is also mentioned by Li et al. (2020).

The performance of the ERA-Int, ERA5, ERA5-L, and HAR datasets in representing the spatial distribution of the precipitation pattern in Nepal in terms of the MB during the summer monsoon season (June-September) when large errors occurred because of a large amount of precipitation, is shown in Fig. 4. In addition, statistical metrics during the monsoon season in the study period were calculated at stations in low-elevation regions (below 1500 m), mid-elevation regions (between 1500 and 2500 m), and high-elevation regions (above 2500 m) (presented in Fig. 5). For the reanalysis datasets, the coarse resolution (0.7°) ERA-Int dataset generally underestimated the amount of precipitation at stations in low-elevation regions (with a gently steep average slope of 17.55°) and mid-elevation regions (with a moderately steep average slope of 27.90°), as indicated by red dots in Fig. 4a and negative MB values in Fig. 5a, while it generally overestimated the precipitation at stations in high-elevation regions (with a very steep average slope of 32.98°), as indicated by black dots in Fig. 4a and positive MB values in Fig. 5a. In contrast with ERA-Int, ERA5 and ERA5-L with the higher resolutions (0.25° and 0.1°, respectively) generally overestimated the precipitation at not only the stations in high-elevation regions but also most of the stations in mid- and low-elevation regions. For the entire study region, the coarse resolution ERA-Int dataset generally underestimated the amount of precipitation with an average MB of −1.37 mm/day and a low R-value (0.27), indicating a low spatial relevance

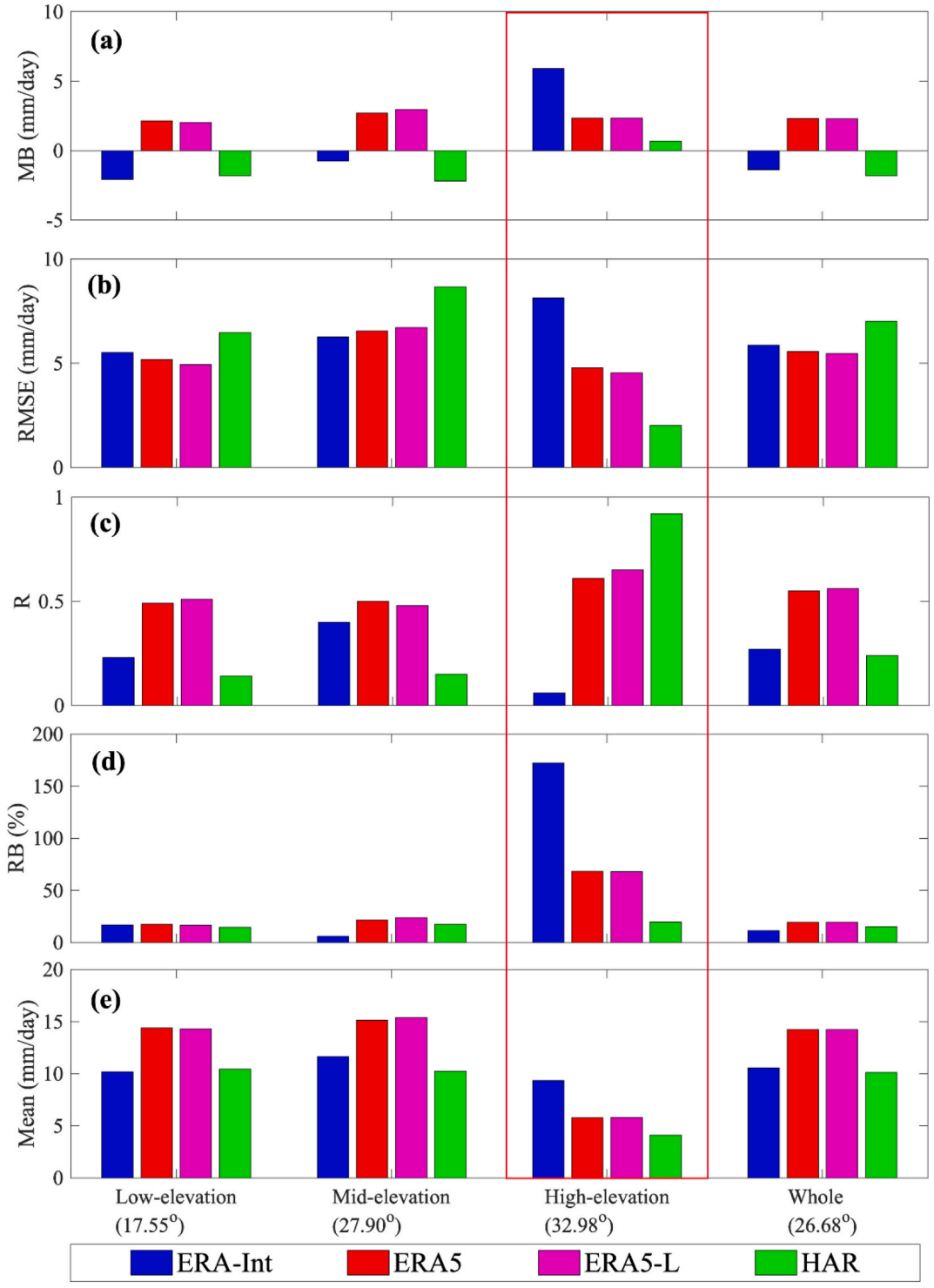



**Fig. 5.** Statistical metrics of four datasets during the summer monsoon season (June-September) derived from the mean monthly precipitation (mm/day) in the study period are calculated for low-elevation (below 1500 m), mid-elevation (between 1500 and 2500 m), and high-elevation (above 2500 m) regions. The higher resolution simulations (ERA5, ERA5–Land, and HAR) outperform ERA-Interim in high–elevation areas as emphasized by the metrics in the red rectangle. The average slope for each elevation range is also given in parentheses. The slope is calculated using 30 m SRTMDEM datasets.

with OBS, which is consistent with the fact revealed by Fig. 3b. The high-resolution ERA5 and ERA5-L datasets generally overestimated the precipitation with similar average MBs of 2.31 and 2.30 mm/day, respectively (Fig. 5a). The R-values of the ERA5 and ERA5-L datasets (i.e., 0.55 and 0.56, respectively) are evidently higher than that of the ERA-Int dataset, indicating a better spatial performance with OBS, which is also consistent with Figs. 3c and d. Because the ERA5-L dataset uses ERA5 atmospheric variables (i.e., temperature and humidity) as forcing data to control the simulated land fields, it shows a very similar spatial distribution with a slightly higher precipitation amount than that of ERA5. The refined HAR dataset (with a spatial resolution of 0.1°) has a similar performance with that of the ERA-Int dataset in low- and mid-elevation regions but a better performance in high-elevation regions with smaller errors and higher R-values (Fig. 5b). The high-resolution

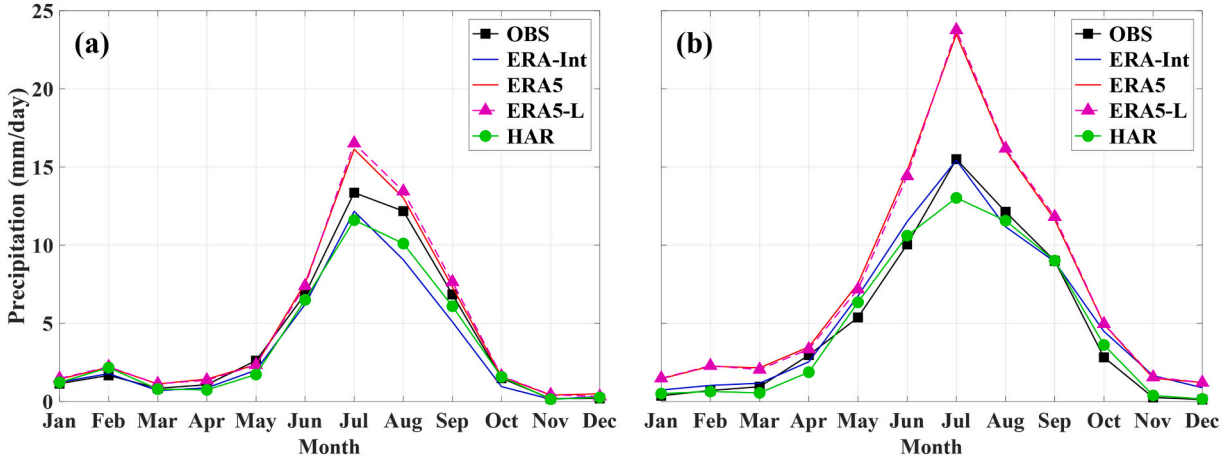



**Fig. 6.** Mean annual precipitation variation in the OBS, ERA-Int, ERA5, ERA5-L, and HAR datasets over two regions, i.e., (a) the western region (80–83°E) and (b) the eastern region (86–88°E), derived from their mean monthly precipitations (mm/day) grouped for the respective regions during 2001–2013.

ERA5 and ERA5-L datasets also performed better than the ERA-Int dataset with smaller errors and higher R-values in high-elevation regions (Fig. 5). These facts suggested that the high-resolution datasets can better present the fine-scale details of the precipitation pattern in high-elevation regions, where the terrain is more complex with a very steep average slope of 32.98°.

### 3.2. Comparison of the annual precipitation variation for Western and Eastern Nepal

The mean annual precipitation variation, as reflected by OBS, reanalysis, and refined datasets, for the western and eastern regions of Nepal is shown in Fig. 6. The highest amount of precipitation occurred in July because of the summer monsoon control. We only show the western and eastern regions in Fig. 6 to demonstrate the difference in the east-west monsoon penetration duration between the two regions. In the western region, precipitation abruptly decreases from August to September (strong monsoon weakening), while in the eastern region, precipitation gradually decreases from August to October (gradual monsoon weakening). All four datasets capture the short monsoon period (May-October) in the western region and the long monsoon period (April-October) in the eastern region, as presented by OBS. On the other hand, the coarse resolution ERA-Int dataset well reproduces the observed seasonal variation range of the precipitation in the eastern region but evidently underestimates it in the west region. In contrast with ERA-Int, the fine resolution ERA5 and ERA5-L datasets perform very similarly, dramatically overestimating the seasonal variation range in the eastern region and slightly overestimating it in the western region. As we know, the water vapor of monsoons in Nepal primarily comes from the Bay of Bengal and transfers from east to west. These facts suggest that the ERA5 and ERA5-L simulations with high-resolution effectively reflect the blocking effects of the terrain and dam more water vapor from the Bay of Bengal in the eastern region, therefore less water vapor is transported into the western region. Similar to the findings in Section 3.1, the refined HAR dataset performs in close agreement with ERA-Int but presents a slightly smaller range in the seasonal variation of precipitation in the two regions.

Fig. 7 provides the statistical metrics for the study period derived from the regional monthly mean precipitations for three regions. The error metrics further confirm the findings derived from Fig. 6. ERA-Int and HAR showed similar smaller MBs, RMSEs, and RBs for the eastern region than for the western region (Fig. 7a, b and d). The high-resolution ERA5 and ERA5-L datasets evidently overestimate the seasonal variation range of precipitation in the eastern region with a larger MB and RMSE of 2.34 mm/day and ~3.45 mm/day, respectively, while they slightly overestimate the seasonal variation range in the western region with smaller errors. All four datasets show high correlation values ($R > 0.90$), indicating that they can well reproduce the seasonal precipitation cycle (but with different variation ranges) over the country (Fig. 7c).

### 3.3. Elevation dependency of precipitation

Knowledge of the elevation gradient of precipitation is very important for many hydrometeorological applications. The highest proportion of precipitation occurs during the summer monsoon season (Fig. 8); thus, the elevation dependency was investigated based on monthly data from the summer monsoon season (JJAS). The mean precipitation data from the OBS, reanalysis, and refined datasets were averaged over the summer monsoon at different elevation ranges of 500 m between 60 m and > 4000 m (Fig. 8) during the study period. Few stations are above 3000 m (shaded rectangle in Fig. 8).

The high mountain located in southern slope of central Himalaya acts as a natural barrier to water vapor transport to the high–lands, the precipitation decreases with increasing elevation, especially above 2000 m. The observed mean monsoonal (JJAS) precipitation shows a clear elevation dependency, i.e., precipitation increasing gradually from ~12 to ~14 mm/day with elevation increasing below 2000 m and then decreasing quickly with increasing elevation (black line in Fig. 8). The highest amount of mean monsoonal precipitation (approximately 14 mm/day) is detected in the 1500–2000 m range; all four reanalysis and refined datasets also detect the maximum amount of precipitation in this range but with different values. The ERA5-L and ERA5 datasets reproduce this elevation dependency but tend to overestimate the precipitation. HAR also reproduces this elevation dependency but tends to underestimate the precipitation. However, ERA-Int does not fully capture the elevation dependency as revealed by the OBS dataset, especially in the high-elevation regions with a very steep average slope of 32.98°. Thus, we speculate that a higher resolution numerical model can more precisely simulate regional climates (e.g., better simulate the spatial pattern of precipitation) by solving more detailed atmospheric patterns and the orographic variance in mountainous regions (Derin and Yilmaz, 2014; Bui et al., 2019).

In general, the high mountainous region on the southern slope of the Himalayas with a steep topography presents a heterogeneous distribution of the precipitation over the study region. The uplift of wet moisture striking against these tall mountains provides more precipitation on the windward side and less precipitation in the leeward region (Fig. 3). On the other hand, the high-resolution data shows more precipitation on the southern slope of the Himalayas because, according to the high-resolution simulation, more water vapor is trapped because of the

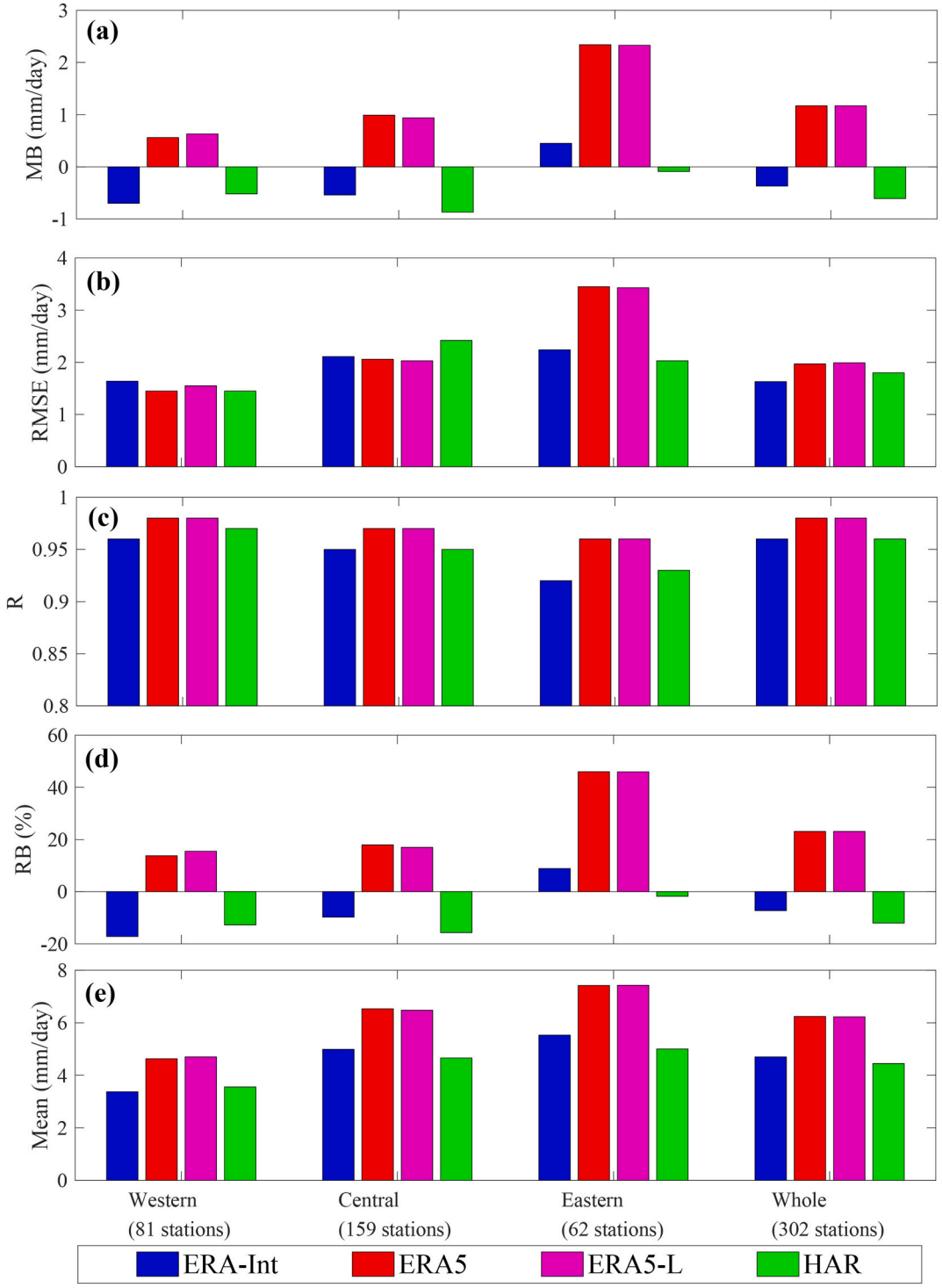



**Fig. 7.** Statistical metrics; (a) the mean bias (MB), (b) RMSE, (c) correlation coefficient (R), (d) relative bias (RB), and (e) mean precipitation in the western, central, and eastern regions, as well as in the entire study region, derived from the regional mean monthly precipitation (mm/day).

roughness related to the steeper land surface. Meanwhile, the coarse resolution data show less precipitation because more water vapor is transported into the Tibetan Plateau because of less roughness related to the smoother land surface. These assumptions are supported by the simulation study conducted on the Tibetan Plateau, China (Wang et al., 2020b), and strongly supported by the statistical analysis of this study.

## 4. Discussion

This study evaluates the spatial performance of the multiple reanalysis and refined precipitation datasets using 302 rain gauge stations from the southern slope of central Himalaya, Nepal. The rugged topographic nature, local climate and model resolution are some of the factors that influence the performances of the reanalysis and refined model simulation estimations.

The spatial distribution of the precipitation is primarily determined

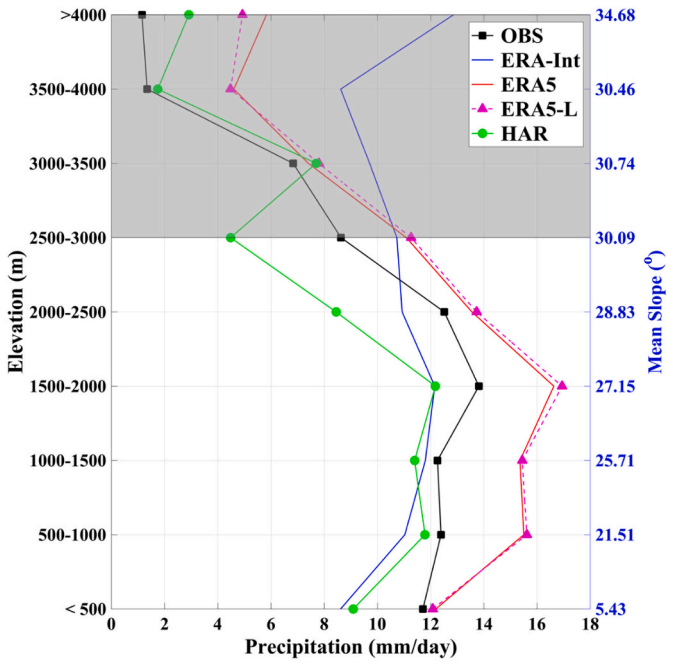



**Fig. 8.** Mean precipitation amount averaged over the summer monsoon season (mm/day) at different elevation ranges along with the average slope during the study period, for OBS, ERA-Int, ERA5, ERA5-L, and HAR at the station locations. The shaded rectangle shows the limited number of stations in that elevation range.

by the interaction of natural controls, such as the latitude, elevation, and slope as well as an orographic influence (Smith and Barstad, 2004; Blocken et al., 2006). The southern slope of central Himalaya is characterized by steep and rugged terrain with ridges and valleys. The mountainous geography and characteristics of the airflow create variation in the distribution of precipitation over the study region (Karki et al., 2017). However, these mountainous regions are one of the complicated areas for meteorological simulations. The estimates of orographic precipitation in reanalysis datasets are affected by the model resolution and other physical processes (Fallah et al., 2019; Xu et al., 2019). The high-resolution datasets can better represent the terrain effect, moisture transport and precipitation process in a mountainous region; thus, ERA5-L and ERA5 show a more consistent spatial pattern of observed precipitation than that of coarse resolution ERA-Int. However, our results show HAR, ERA5-L, and ERA5 outperform ERA-Int only in high-elevation regions (above 3000 m); in low-elevation regions they all perform very similarly. Furthermore, the large discrepancy in the ERA-Int dataset was more pronounced in high-elevation regions. In these regions, coarse resolution simulations smoothen the land surface, while high-resolution simulations present a steeper and rougher land surface similar to that of areal terrain; therefore, they better represent topography-induced effects and orographic precipitation (Wang et al., 2020b).

The reanalysis and refined datasets well capture the monthly variation in precipitation over the western and eastern regions of the country. The monsoon season contributes 80% of the annual precipitation in Nepal (Sharma et al., 2020); in this season, water vapor primarily comes from the Bay of Bengal, and transfers from east to west. ERA5-L and ERA5 largely overestimated the monsoonal precipitation in the eastern region, in contrast to the western region, suggesting that the ERA5 and ERA5-L simulations with high-resolution effectively reflect the blocking effects of the terrain and dam more water vapor from the Bay of Bengal in the eastern region, therefore allowing less water vapor transport into the western region. Moreover, the higher spatial resolution and improved physical and process representations could explain the improved performances of ERA5-L and ERA5 over ERA-Int (Tarek et al., 2019). It is worth mentioning that ERR5-L uses ERA5 atmospheric variables (i.e., temperature and humidity) as forcing data to control the simulated land fields; however, they both perform very similarly. This result indicates that, without further data enhancements, a slightly higher spatial resolution might be useful for regions with a steep topography but may not lead to performance improvements (Zandler et al., 2019). Meanwhile, most of the DHM stations in high-elevation regions are located in valley bottoms (Lang and Barros, 2004; Ichiyanagi et al., 2007), where the slope and wind circulation have a remarkable effect that may result in an underestimation of the observed precipitation (Barry, 1992). In such regions, these OBS data may not capture the representative precipitation variability, and the uncertainty leaves a gap in the quantification of gridded reanalysis and refined datasets.

In fact, precipitation is not directly assimilated in the reanalysis and refined datasets; however, some observations, such as the temperature and humidity, are used to initialize the forecast model; hence, the accuracy of the model simulation primarily depends on the quality of the model physics (Zhang et al., 2013). The precipitation datasets from reanalysis (ERA-Int, ERA5, and ERA5-L) and the regional refinement (HAR) that are used in the study are forecast variables. Therefore, the abilities of the above models to simulate precipitation over Nepal are highly dependent on their abilities to express the dynamical and thermal effects of complex terrain at different horizontal scales (Ma et al., 2009; Palazzi et al., 2013), which could be associated with the dynamical core, sub-grid scale physical parameterizations and the horizontal resolutions during model configuration. For example, the sub-grid scale orographic drag scheme directly modulates the atmospheric circulation and local land-atmosphere interactions (Zhang et al., 2006; Zhou et al., 2017; Zhou et al., 2018); the cumulus convection scheme directly modifies the convective precipitation within one grid cell. Thus, high-resolution simulations can resolve more orography and yield better simulation results than low-resolution simulations (Lin et al., 2018). These differences in the models used by the different reanalysis and refined datasets are likely the main reasons for uncertainties in precipitation forecasting. There are also potential errors due to the comparison between the observed gauge point measurement and the model grid, which is the average of a grid box measuring 0.75°, 0.25°, 0.1°, and 0.1° for ERA-Int, ERA5, ERA5-L, and HAR, respectively. For example, grid boxes with two or more rain gauges could have differences in gauge measurements that can be attributed to the generally high spatio-temporal precipitation data (Nkiaka et al., 2017).

## 5. Summary

This study evaluates the spatial performances of three reanalysis precipitation datasets (ERA-Int, ERA5, and ERA5-L) from the ECMWF and the refined HAR precipitation product downscaled using the WRF model, with the spatial resolution ranging from coarse to fine (0.7°, 0.25°, 0.1° and 0.1°, respectively), from 2001 to 2013 using 302 rain gauge observations on the southern slope of central Himalaya, Nepal.

The high mountains region on the southern slope of the Himalayas with steep topography presents a heterogeneous distribution of the precipitation over the study region. The uplift of wet moisture striking against these high mountains provides more precipitation on the windward side and less precipitation in the leeward region of the country. The high-resolution simulations reflect the damming of more water vapor because of more rough land surface related with the steeper land surface and show a consistent spatial pattern of precipitation (with the highest correlation coefficient), as observation. Meanwhile, the coarse resolution data show less precipitation because they allow more water vapor transport into the Tibetan Plateau because of less roughness related to the smoother land surface.

The monsoon water vapor enters Nepal from the east and transfers toward the west, and the high-resolution simulations tend to effectively

resemble the true precipitation in the eastern region because they reflect the blocking effects of the terrain and dam more water vapor in the eastern and central region. In contrast, coarse resolution datasets do not resemble true precipitation because they allow more water vapor transport into the western region as they cannot capture terrain blocking effects.

All four datasets perform differently for different elevation ranges. The high-resolution datasets show a similar elevation dependency of precipitation with OBS and outperform the coarse-resolution datasets, especially in high-elevation regions, because high-resolution datasets can better present the effects of an orographic gradient on precipitation simulation in mountainous regions. All four datasets perform very similarly in low-elevation regions because of the very smooth topography and the less pronounced effect of orography.

The present work quantifies the spatial performances of the ERA-Int, ERA5, ERA5-L, and HAR datasets on the southern slope of the central Himalayas in Nepal. This evaluation delivers a statistical basis and provides a concrete outlook on data selection in meteorological, hydrological, glaciological, and disaster-related studies within the study region.

## Declaration of Competing Interest

None.

*Acknowledgements*

We would like to thank the ECMWF and Chair of Climatology, TU Berlin, for the public access of their data via their webpages. The Department of Hydrology and Meteorology (DHM), the government of Nepal, is also acknowledged for providing the observed precipitation datasets. This study was jointly supported by the Second Tibetan Plateau Scientific Expedition and Research Program (STEP) (2019QZKK0206), the National Natural Science Foundation of China (41871280 and 41471286), and the 13th five-year Informatization Plan of Chinese Academy of Sciences (XXH13505-06).

## References

Albergel, C., Dutra, E., Munier, S., Calvet, J.-C., Munoz-Sabater, J., de Rosnay, P., Balsamo, G., 2018. ERA-5 and ERA-Interim driven ISBA land surface model simulations: which one performs better? Hydrol. Earth Syst. Sci. 22.

Barros, A.P., Lang, T.J., 2003. Monitoring the monsoon in the Himalayas: Observations in Central Nepal, June 2001. Mon. Weather Rev. 131, 1408–1427.

Barros, A.P., Chiao, S., Lang, T.J., Burbank, D., Putkonen, J., 2006. From weather to climate-Seasonal and interannual variability of storms and implications for erosion processes in the Himalaya. Geol. Soc. Am. Spec. Pap. 398, 17–38.

Barry, R.G., 1992. Mountain climatology and past and potential future climatic changes in mountain regions - a review. Mt. Res. Dev. 12, 71–86.

Bhardwaj, A., Ziegler, A.D., Wasson, R.J., Chow, W.T., 2017. Accuracy of rainfall estimates at high altitude in the Garhwal Himalaya (India): a comparison of secondary precipitation products and station rainfall measurements. Atmos. Res. 188, 30–38.

Blocken, B., Poesen, J., Carmeliet, J., 2006. Impact of wind on the spatial distribution of rain over micro-scale topography: numerical modelling and experimental verification. Hydrol. Process. 20, 345–368.

Bosilovich, M.G., Kennedy, J., Dee, D., Allan, R., O'Neill, A., 2013. On the Reprocessing and Reanalysis of Observations for Climate. Climate Science for Serving Society. Springer, pp. 51–71.

Bui, H.T., Ishidaira, H., Shaowei, N., 2019. Evaluation of the use of global satellite–gauge and satellite-only precipitation products in stream flow simulations. Appl Water Sci 9, 53.

C3S, 2017. Copernicus Climate Change Service: ERA5: Fifth Generation of ECMWF Atmospheric Reanalyses of the Global Climate. Copernicus Climate Change Service Climate Data Store (CDS). November, 2019. https://cds.climate.copernicus.eu/cdsapp#!/home.

C3S, 2019. C3S ERA5-land Reanalysis. Copernicus Climate Change Service. date of access December, 2019. https://cds.climate.copernicus.eu/cdsapp#!/home.

Ceglar, A., Toreti, A., Balsamo, G., Kobayashi, S., 2017. Precipitation over monsoon Asia: a comparison of reanalyses and observations. J. Clim. 30, 465–476.

Dee, D.P., et al., 2011. The ERA-Interim reanalysis: Configuration and performance of the data assimilation system. Q. J. R. Meteorol. Soc. 137, 553–597.

Delalay, M., Ziegler, A.D., Shrestha, M.S., Wasson, R.J., Sudmeier-Rieux, K., McAdoo, B. G., Kochhar, I., 2018. Towards improved flood disaster governance in Nepal: a case study in Sindhupalchok District. Int. J. Disaster Risk Reduct. 31, 354–366.

Derin, Y., Yilmaz, K.K., 2014. Evaluation of multiple satellite-based precipitation products over complex topography. J. Hydrometeorol. 15, 1498–1516.

DHM, 2017. Observed Climate Trend Analysis in the Districts and Physiographic Regions of Nepal (1971–2014), 1. Department of Hydrology and Meteorology, Kathmandu, p. 74.

Essou, G.R.C., Brissette, F., Lucas-Picher, P., 2017. The use of reanalyses and gridded observations as weather input data for a hydrological model: comparison of performances of simulated river flows based on the density of weather stations. J. Hydrometeorol. 18, 497–513.

Faiz, M.A., et al., 2018. How accurate are the performances of gridded precipitation data products over Northeast China? Atmos. Res. 211, 12–20.

Fallah, A., Rakhshandehroo, G.R., Berg, P., O, S., Orth, R., 2019. Evaluation of precipitation datasets against local observations in Southwestern Iran. Int. J. Climatol. 40 (9), 4102–4116. https://doi.org/10.1002/joc.6445.

Feidas, H., 2010. Validation of satellite rainfall products over Greece. Theor. Appl. Climatol. 99, 193–216.

Goodison, B.E., Louie, P.Y., Yang, D., 1998. WMO Solid Precipitation Measurement Intercomparison.

Hamal, K., et al., 2020. Evaluation of MERRA-2 precipitation products using gauge observation in Nepal. Hydrology 7, 40. https://doi.org/10.3390/hydrology7030040.

Hamal, K., Sharma, S., Baniya, B., Khadka, N., Zhou, X., 2020. Inter-annual variability of Winter Precipitation over Nepal coupled with ocean-atmospheric patterns during 1987–2015. Front. Earth Sci. 8, 161. https://doi.org/10.3389/feart.2020.00161.

Hersbach, H., Dee, D., 2016. ERA5 reanalysis is in production. ECMWF Newslett. 147.

Hu, Z.Y., Hu, Q., Zhang, C., Chen, X., Li, Q.X., 2016. Evaluation of reanalysis, spatially interpolated and satellite remotely sensed precipitation data sets in Central Asia. J. Geophys. Res.-Atmos. 121, 5648–5663.

Ichiyanagi, K., Yamanaka, M.D., Muraji, Y., Vaidya, B.K., 2007. Precipitation in Nepal between 1987 and 1996. Int. J. Climatol. 27, 1753–1762.

Jiang, S., Ren, L., Hong, Y., Yong, B., Yang, X., Yuan, F., Ma, M., 2012. Comprehensive evaluation of multi-satellite precipitation products with a dense rain gauge network and optimally merging their simulated hydrological flows using the Bayesian model averaging method. J. Hydrol. 452, 213–225.

Kalnay, E., et al., 1996. The NCEP/NCAR 40-year reanalysis project. Bull. Am. Meteorol. Soc. 77, 437–471.

Kansakar, S.R., Hannah, D.M., Gerrard, J., Rees, G., 2004. Spatial pattern in the precipitation regime of Nepal. Int. J. Climatol. 24, 1645–1659.

Karki, R., Talchabhadel, R., Aalto, J., Baidya, S.K., 2016. New climatic classification of Nepal. Theor. Appl. Climatol. 125, 799–808.

Karki, R., Schickhoff, U., Scholten, T., Böhner, J., 2017. Rising precipitation extremes across Nepal. Climate 5, 4.

Kidd, C., Huffman, G., 2011. Global precipitation measurement. Meteorol. Appl. 18, 334–353.

Kolluru, V., Kolluru, S., Konkathi, P., 2020. Evaluation and integration of reanalysis rainfall products under contrasting climatic conditions in India. Atmos. Res. 246, 105121.

Lang, T.J., Barros, A.P., 2002. An investigation of the onsets of the 1999 and 2000 monsoons in Central Nepal. Mon. Weather Rev. 130, 1299–1316.

Lang, T.J., Barros, A.P., 2004. Winter storms in the Central Himalayas. J. Meteorol. Soc. Jpn. Ser. II 82, 829–844.

Li, Z., Yang, D., Hong, Y., 2013. Multi-scale evaluation of high-resolution multi-sensor blended global precipitation products over the Yangtze River. J. Hydrol. 500, 157–169.

Li, H., Haugen, J.E., Xu, C., 2017. Precipitation pattern in the western himalayas revealed by four datasets. In: Hydrology and Earth System Sciences Discussions, pp. 1–19.

Li, D., Yang, K., Tang, W., Li, X., Zhou, X., Guo, D., 2020. Characterizing precipitation in high altitudes of the western Tibetan plateau with a focus on major glacier areas. Int. J. Climatol. 1.

Lin, C.G., Chen, D.L., Yang, K., Ou, T.H., 2018. Impact of model resolution on simulating the water vapor transport through the Central Himalayas: implication for models' wet bias over the Tibetan Plateau. Clim. Dyn. 51, 3195–3207. https://doi.org/10.1007/s00382-018-4074-x.

Lindsay, R., Wensnahan, M., Schweiger, A., Zhang, J., 2014. Evaluation of seven different atmospheric reanalysis products in the Arctic*. J. Clim. 27, 2588–2606.

Liu, X., Yang, T., Hsu, K., Liu, C., Sorooshian, S., 2017. Evaluating the streamflow simulation capability of PERSIANN-CDR daily rainfall products in two river basins on the Tibetan Plateau. Hydrol. Earth Syst. Sci. 21, 169.

Lu, D., Yong, B., 2018. Evaluation and hydrological utility of the latest GPM IMERG V5 and GSMaP V7 precipitation products over the Tibetan Plateau. Remote Sens. 10, 2022.

Ma, L., Zhang, T., Frauenfeld, O.W., Ye, B., Yang, D., Qin, D., 2009. Evaluation of precipitation from the ERA-40, NCEP-1, and NCEP-2 reanalyses and CMAP-1, CMAP-2, and GPCP-2 with ground-based measurements in China. J. Geophys. Res. 114.

Maussion, F., Scherer, D., Molg, T., Collier, E., Curio, J., Finkelnburg, R., 2014. Precipitation seasonality and variability over the Tibetan Plateau as resolved by the high Asia reanalysis. J. Clim. 27, 1910–1927.

Menegoz, M., Gallee, H., Jacobi, H.W., 2013. Precipitation and snow cover in the Himalaya: from reanalysis to regional climate simulations. Hydrol. Earth Syst. Sci. 17, 3921–3936.

Nayava, J.L., 1980. Rainfall in Nepal. Himal. Rev. 12, 1–18.

Nkiaka, E., Nawaz, N., Lovett, J., 2017. Evaluating global reanalysis precipitation datasets with rain gauge measurements in the Sudano-Sahel region: case study of the Logone catchment, Lake Chad Basin. Meteorol. Appl. 24, 9–18.
Palazzi, E., von Hardenberg, J., Provenzale, A., 2013. Precipitation in the Hindu-Kush Karakoram Himalaya: observations and future scenarios. J. Geophys. Res.-Atmos. 118, 85–100.
Pokharel, B., Wang, S.Y.S., Meyer, J., Marahatta, S., Nepal, B., Chikamoto, Y., Gillies, R., 2019. The east–west division of changing precipitation in Nepal. Int. J. Climatol. 40, 3348–3359.
Rana, S., McGregor, J., Renwick, J., 2015. Precipitation seasonality over the Indian subcontinent: an evaluation of gauge, reanalyses, and satellite retrievals. J. Hydrometeorol. 16, 631–651.
Sahlu, D., Moges, S.A., Nikolopoulos, E.I., Anagnostou, E.N., Hailu, D., 2017. Evaluation of high-resolution multisatellite and reanalysis rainfall products over East Africa. Adv. Meteorol. 2017.
Sarojini, B.B., Stott, P.A., Black, E., 2016. Detection and attribution of human influence on regional precipitation. Nat. Clim. Chang. 6, 669–675.
Sharma, S., et al., 2020. Evaluation of GPM-Era satellite precipitation products on the southern slopes of the Central Himalayas against rain gauge data. Remote Sens. 12, 1836. https://doi.org/10.3390/rs12111836.
Sharma, S., Hamal, K., Khadka, N., Joshi, B.B., 2020. Dominant pattern of year-to-year variability of summer precipitation in Nepal during 1987–2015. Theor. Appl. Climatol. 140. https://doi.org/10.1007/s00704-020-03359-1.
Shrestha, M.L., 2000. Interannual variation of summer monsoon rainfall over Nepal and its relation to Southern Oscillation Index. Meteorog. Atmos. Phys. 75, 21–28.
Smith, R.B., Barstad, I., 2004. A linear theory of orographic precipitation. J. Atmos. Sci. 61, 1377–1391.
Sun, Q.H., Miao, C.Y., Duan, Q.Y., Ashouri, H., Sorooshian, S., Hsu, K.L., 2018. A review of global precipitation data sets: data sources, estimation, and intercomparisons. Rev. Geophys. 56, 79–107.
Talchabhadel, R., Karki, R., Parajuli, B., 2017. Intercomparison of precipitation measured between automatic and manual precipitation gauge in Nepal. Measurement 106, 264–273.
Talchabhadel, R., Karki, R., Thapa, B.R., Maharjan, M., Parajuli, B., 2018. Spatio-temporal variability of extreme precipitation in Nepal. Int. J. Climatol. 38, 4296–4313.
Tapiador, F.J., et al., 2012. Global precipitation measurement: methods, datasets and applications. Atmos. Res. 104, 70–97.
Tarek, M., Brissette, F.P., Arsenault, R., 2019. Evaluation of the ERA5 reanalysis as a potential reference dataset for hydrological modeling over North-America. Hydrol. Earth Syst. Sci. Discuss. 1–35.
Trenberth, K.E., Dai, A., Rasmussen, R.M., Parsons, D.B., 2003. The changing character of precipitation. Bull. Am. Meteorol. Soc. 84, 1205–1217.
van der Geest, K., 2018. Landslide loss and damage in Sindhupalchok district, Nepal: comparing income groups with implications for compensation and relief. Int. J. Disaster Risk Sci. 1–10.
Wang, Y., et al., 2020a. Synergy of orographic drag parameterization and high resolution greatly reduces biases of WRF-simulated precipitation in central Himalaya. Clim. Dyn. 54, 1729–1740.
Wang, N., Liu, W., Sun, F., Yao, Z., Wang, H., Liu, W., 2020b. Evaluating satellite-based and reanalysis precipitation datasets with gauge-observed data and hydrological modeling in the Xihe River Basin, China. Atmos. Res. 234, 104746.
Xu, X., Frey, S.K., Boluwade, A., Erler, A.R., Khader, O., Lapen, D.R., Sudicky, E., 2019. Evaluation of variability among different precipitation products in the Northern Great Plains. J. Hydrol. 24, 100608.
Yang, F., et al., 2017. Evaluation of multiple forcing data sets for precipitation and shortwave radiation over major land areas of China. Hydrol. Earth Syst. Sci. 21, 5805–5821.
Yao, J., Chen, Y., Yu, X., Zhao, Y., Guan, X., Yang, L., 2020. Evaluation of multiple gridded precipitation datasets for the arid region of northwestern China. Atmos. Res. 236, 104818.
Zandler, H., Haag, I., Samimi, C., 2019. Evaluation needs and temporal performance differences of gridded precipitation products in peripheral mountain regions. Sci. Rep. 9, 1–15.
Zhang, Y.C., Huang, A.N., Zhu, X.S., 2006. Parameterization of the thermal impacts of sub-grid orography on numerical modeling of the surface energy budget over East Asia. Theor. Appl. Climatol. 86, 201–214.
Zhang, Q., Körnich, H., Holmgren, K., 2013. How well do reanalyses represent the southern African precipitation? Clim. Dyn. 40, 951–962.
Zhou, X., Beljaars, A.C.M., Wang, Y., Huang, B., Lin, C., Chen, Y., Wu, H., 2017. Evaluation of WRF simulations with different selections of sub-grid orographic drag over the Tibetan Plateau. J. Geophys. Res.-Atmos. 122, 7755–7771.
Zhou, X., Yang, K., Wang, Y., 2018. Implementation of a turbulent orographic form drag scheme in WRF and its application to the Tibetan Plateau. Clim. Dyn. 50, 2243–2255.